MIL-S-23891C

30 January 1985
SUPERSEDING
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29 November 1972

MILITARY SPECIFICATION

SPRAYER-DUSTER, BACKPACK TYPE; GED BLOWER (COMMERCIAL)

This specification is approved for use by all Departments and Agencies of the Department of the Defense.

1. SCOPE

1.1 Scope: This specification covers a lightweight backpack dispersal device, gasoline-engine-driven (GED), mist-dust blower.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	-	Treatment and Painting of Materiel.
MIL-S-12651	-	Sprayers, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Medical Bioengineering Research and Development Laboratory, ATTN: SGRD-UBE-V, Fort Detrick, MD 21701-5010, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3740

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STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-882	-	System Safety Program Requirements.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474	-	Noise Limits for Army Materiel.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Qualification. The sprayer furnished under this specification shall be a product that has been tested and has passed the qualification inspection specified herein; and has been listed on, or approved for listing on, the applicable qualified products list. One or more sprayers, as specified by the Government, shall be presented by the manufacturer for qualification inspection.

3.2 Description. The sprayer-duster shall consist of a gasoline-engine-driven air blower, an air agitated dispersant reservoir, a backpack type mount, air and dispersant hoses (minimum 2 feet long) and nozzle, and all accessories for safety and satisfactory operation of the unit while being carried on the back of an operator. The machine shall be designed to disperse pesticide liquids, granules, and dusts. The horizontal throw distance for each dispersant shall be at least 20 feet. A grounding chain that effectively eliminates static shock to the operator when dispersing shock producing formulations shall be provided. The assembly shall have a maximum unfilled weight less than 30 pounds.

3.3 Material. Material shall be as specified herein. Material not definitely specified shall be of the best quality used for the purpose in commercial practice, free from all defects and imperfections that might affect the serviceability of the finished product. Material used in the engine mounts shall be corrosion resistant. Seals, O-rings, or any insecticide conveying component shall be made of nylon, polyethylene, viton, polytetra-fluoroethylene, or other suitable corrosion-resistant material.

3.4 Interchangeability. All units of the same type and description shall be functionally and dimensionally interchangeable. This requirement includes parts, assemblies, and accessories. No deviation shall be acceptable without prior written approval from the contracting officer.

3.5 Standard product. Except where modified herein, it is intended that the equipment and its component parts shall be a regular commercial product of the manufacturer or his suppliers. The parts shall be new and unused.

3.6 Performance. The sprayer-duster units shall be designed to disperse insecticide formulations of oil and water based solutions, emulsions, wettable powders, dusts, and granules.

3.6.1 Reliability. The sprayer-duster shall be designed to satisfy the criteria of 4.5.1 and 4.5.2 at the speed required for delivery of liquids, dusts, or granules at the maximum flow rate specified for the sprayer.

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3.6.2 Liquid, dust, and granule dispersal rate control. The sprayer-duster shall be designed with convenient dispersal rate controls for the three dispersants, i.e., liquids, granules, and dusts. The controls shall be capable of regulating flow of liquids to a minimum range of 0.1 to 2.0 gallons/hour, dusts over a minimum range of 50 to 120 pounds/hour, and granules to a minimum of 12 pounds/hour. Each control shall be continuous or stepped in at least three increments. Flow controls shall remain fixed in position once set by the operator.

3.6.3 Maintainability. The sprayer-duster shall be designed to satisfy the maintainability criteria in 4.5.3.

3.7 Design. The equipment design and accessory installations shall permit accessibility for maintenance and service in the field. The design shall be such as to prevent conditions that may be hazardous to personnel or deleterious to equipment, and shall comply with safety provision the guidelines of MIL-STD-882.

3.8 Gasoline engine. The gasoline engine shall be a heavy-duty, lightweight, air-cooled type with a continuous horsepower rating sufficient to meet the performance requirements but with not less than 30 cc of displacement. The engine furnished shall produce the required sprayer performance on unleaded or low-lead content gasoline. The content of antiknock compounds (metallic lead) for low-lead gasoline shall not exceed 0.5 gram/gallon. The fuel-to-oil mix ratio of two-cycle engines shall be clearly labeled on the machine. The engine shall be of a standard commercial design that has been proven to be satisfactory in extensive use and shall be equipped with a recoil-type pull-rope starter with a starting sheave positioned clear of obstruction. The air cooling system shall be such that the engine will operate satisfactorily over long periods (at least 20 hours) when operating the equipment at full load in an ambient atmosphere of 100°F. The lubrication system shall provide adequate lubrication under all operating conditions. A washable foam-type air filter shall be provided directly on the carburetor air intake. Carburetor setting screws shall be readily accessible for adjustments without requiring removal of any parts or without use of offset-type tools.

3.9 Exhaust system. The engine exhaust pipe shall be provided with a muffler with the exhaust directed away from the operator. The exhaust pipe mounting shall be designed for easy removal within 10 man-minutes using only common hand tools in the service kit provided with the machine. Any exposed parts of the exhaust system shall be located to minimize heat or danger of burns to the operator.

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3.10 Fuel tank. A fuel tank with a capacity sufficient for 1/2 hour normal operation shall be provided. A strainer (which is easily removed for cleaning) shall be provided on the line from the fuel tank to the carburetor. The filler cap on the fuel tank shall be mounted as far from the exhaust outlets as possible, but with easy access for filling. The fuel tank shall be designed with a neck and filler cap such that no fuel will seep through the vent of the cap when the engine is operating at normal rpm and the tank is filled to the bottom of the neck. If a gravity fuel tank is used, an easily accessible shutoff valve shall be provided.

3.11 Noise limits. The noise level of the sprayer assembly shall not exceed 85 decibels on an A-weighted scale (dB(A)) when measured at points equal to the locations of the ears of an operator. The level of 95 dB(A) may be substituted for the 85 dB(A) sound level requirement if analysis demonstrates that obtaining 85 dB(A) or less is beyond the state-of-the-art, cost prohibitive, or that system effectiveness is seriously degraded by reducing the noise level to 85 dB(A). The level of 95 dB(A) shall not be exceeded without written approval of the procuring activity.

3.12 Blower. The blower fan shall be of the axial or centrifugal type, constructed of lightweight, corrosion-resistant metal. A large mesh debris screen shall be provided on the air intake. To prevent ingestion of foreign material, the intake for the air blower shall be positioned behind the backpad at a minimum distance of 9 inches above the ground (when the machine is resting on the ground in a standing position). The average dispersal air stream velocity produced by the blower shall be a minimum of 150 miles/hour at the outlet when measured with an anemometer. The minimum size of the outlet shall be 2 inches in diameter, and the horizontal throw distance of each dispersant shall be at least 20 feet.

3.13 Dispersant reservoir. The dispersant reservoir shall be translucent and have a capacity of at least 2.5 gallons. The tank shall display graduation markings in both liters and quarts. It shall be constructed of high density (linear type) polyethylene plastic of sufficient thickness to withstand the operational stresses without cracking or splitting. The reservoir shall be equipped with an air agitation system designed to direct and force the dust or granules through the discharge port. The location of the dust or granule discharge port shall be at the center of an entry section with wall slope and entry width adequate to prevent bridging or packing of dust or granules. Also, the slope shall be sufficient to provide smooth feed of dispersant. The maximum permissible residual of dispersant (after feed ceases) is 10 percent by volume. All interior surfaces shall be smooth and designed with gradual transition of surfaces to facilitate an even flow of dust. All fittings on the reservoir shall be integrally molded with the reservoir body. All dispersant

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reservoir fittings (including fill cap) shall be secured such that pesticide spills are minimized. Cantilevered tube or hose attachment designs shall not be permitted. The design shall be such that tubes or hoses are not convenient lifting handles. The reservoir tank shall be designed for ease of conversion from liquids to granules to dusts. Conversion without parts interchange (valves instead) is preferred. If parts interchange is necessary, the maximum permissible time for conversion shall be 10 man-minutes by an unskilled operator, using only the tools provided in the service kit.

3.14 Dispensing hose and nozzle. The blower discharge hose shall be a 2-foot minimum length and shall be capable of verticle dispersal of liquids, granules, and dusts. The dispersing nozzle shall be securely clamped to or integrally molded with the blower discharge hose. An automatic dispersal shut-off control should be incorporated on the hose. The liquid discharge venturi should be centered in the air stream produced by the blower. The liquid flow tube shall be resistant to chemicals and insecticides. All hose and nozzle components shall comply with 3.3.

3.15 Carrying device. The dispersal device shall be mounted on an adjustable, form-fitting, backpack (hip pack). The weight of the components (empty and full) shall be equally distributed laterally and with the center of gravity of the load as close as possible to the operator's spine at the waistline. The hip pack shall be made of molded fiberglass, reinforced plastic, or formed or stamped aluminum. The structural strength of the material shall be sufficient to keep the backpad rigid and erect when the reservoir tank is fully loaded with liquid. The backpad shall be padded with an insulation of flexible plastic foam, at least 1 inch thick, to protect the operator from vibration of the engine. All padding shall be covered with a material that is resistant to fuel, insecticides, and other chemicals. All weight bearing straps contacting the operator shall be padded or designed to allow the operator to operate the backpack sprayer-duster without undue discomfort.

3.16 Service kits. A service kit shall be provided with each sprayer. The kit shall contain the following parts:

a. Wheel puller, if required for routine maintenance.

b. Tool kit and special tools to perform all scheduled and unscheduled maintenance identified in operator's manual.

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3.17 Cleaning, treatment, and painting. Surfaces normally painted shall be cleaned, treated, and painted as specified herein. Unless otherwise specified (see 6.2.1), the color of the finish coat shall be the manufacturer's standard color.

3.17.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil, grease, welding slag and spatter, all loose mill scale, products of corrosion, dirt, or other foreign substances.

3.17.2 Treatment. As soon as practicable after cleaning, and before any evidence of rust or other contamination can result, metal surfaces shall be treated with a primer pretreatment coating to increase the adhesion of the coating system. The equipment shall be treated in accordance with MIL-T-704.

3.17.3 Painting. For metal surfaces, painting shall consist of at least one coat of rust-inhibiting primer and one coat of finish enamel. The rust-inhibiting primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes except that the total dry film thickness shall be not less than 2.5 mils. The paint shall be free from runs, sags, or other defects. The equipment shall be painted in accordance with MIL-T-704.

3.18 Instruction labels. Each dispersal unit shall be labeled to include the following:

a. Fuel-to-oil mix ratio.

b. Noise hazard caution sign (caution sign shall comply with MIL-STD-147 if 85 dB(A) are exceeded).

c. Brief listing of scheduled maintenance items including frequency recommendations that include, as a minimum, the gap setting for spark plug and 50-hour inspection requirement for carbon buildup in the exhaust port and muffler.

3.19 Operator's and maintenance manuals. Manuals shall conform to applicable style, format, and content appropriate for technical manuals required by the military service.

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3.20 Optional item. When required (see 6.2), the following high mortality items shall be specified:

a. Nozzles.

b. Gaskets.

c. Bands and clamps.

d. O-rings.

e. Hoses.

f. Filters.

g. Spark plug.

h. Contact points.

i. Condenser.

j. Other parts which are identified as high mortality items.

3.21 Workmanship. A visual examination will be conducted to determine if the backpack sprayer-duster is clean and free from sand, dirt, fins, pits, scale, rust, sharp edges, cracks, poor or unsound welds, or any other defect that could impair its performance or serviceability.

3.21.1 Metal fabrication. Metal used in the fabrication of the unit shall be free from kinks or sharp bends. The forming of the material shall be done by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately.

3.21.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have burrs removed. Jamb or locknuts shall be provided, and all bolts, nuts, and screws shall be tight.

3.21.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall develop 100 percent of parent metal strength in the parts connected. Welding shall be in accordance with a welding code acceptable to the service concerned.

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3.21.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

c. Packaging inspection (see 4.7).

4.2.1 Qualification inspection. Qualification inspection shall be performed on one or more sprayers of each size specified. This inspection shall include the examination of 4.4 and the testing of 4.5. Failure of the sprayer to pass the examination or tests shall be cause for rejection. The sprayer may be a standard production unit from the contractor's current inventory provided the unit meets the requirements of the specification and is representative of the manufacturing technique applicable to the remaining units to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample selected in accordance with 4.3. This inspection shall include the examination of 4.4, and the test and rejection criteria of 4.5.

4.3 Sampling. A random sample of sprayers shall be selected from each lot in accordance with MIL-STD-105.

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4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level of 2.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-3.

4.4 Examination. Each sprayer-duster shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Quality conformance tests.

4.5.1 Reliability tests. The test unit shall be filled with fuel (regular gasoline mixed with oil as required) and water (in reservoir tank), started up with the pull rope, set for a minimum dispersal rate, and then operated at the manufacturer's recommended speed continuously until refueling is necessary (minimum 30 minutes). The cycle shall be repeated until a total of 200 hours of reliability testing is completed. During this testing, 91 percent technical grade malathion or mineral oil will be substituted for water so that during each 200 hours of reliability testing, at least 20 hours will involve the application of 91 percent technical grade malathion or mineral oil. Minimum acceptable performance for each test unit is four or less failures (not including catastrophic). In addition to the overall limit of four failures, each test machine (in accordance with the definitions that follow) shall not experience (a) any catastrophic failures, (b) more than one major failure, or (c) more than four minor failures. Note that a major failure cascades into the minor failure category; that is, a major failure would reduce the allowed number of minor failures by one. The Government may require the manufacturer to make minor modifications to correct the cause of any failure identified in the reliability testing, or the manufacturer may volunteer modifications to prevent failure reoccurrence during testing. Any such modifications are subject to Government approval and shall be accomplished at no cost to the Government. Also, the Government shall determine if any tests need to be repeated after modifications are accomplished. The following definitions apply to reliability failure criteria:

a. General definition for failure - Any malfunction that causes inability to commence operation, cessation of operation, degradation below the operating conditions required to continue the test, or a safety hazard to personnel.

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b. Catastrophic failure - A serious failure that totally disables the machine and cannot be repaired in less than 8 man-hours and/or requires part replacements that exceed 50 percent of the cost of the machine.

c. Major failure - A failure that requires 1 to 8 man-hours to repair and/or part replacements that cost between 10 percent and 50 percent of the machine cost.

d. Minor failure - A failure that can be corrected in less than 1 man-hour with the use of common equipment and requires part replacements that do not exceed 10 percent of machine cost. In addition, there are two special categories of minor failures: (1) A starting failure is defined as anytime the machine requires 10 or more pull starts to successfully start the engine; (2) A carbon buildup failure is defined as 25 percent or greater blockage of the exhaust port (determined by visual estimate) after 100 hours or less of testing. (Note that, as a matter of routine maintenance during testing, the exhaust ports will be cleaned after 100 hours--failure at that time will not be noted unless blockage exceeds 25 percent.)

e. Repair time - Repair time does not include diagnostic time; it is intended to reflect the time required by a skilled mechanic familiar with the machine and equipped with all necessary tools and parts. Fruitless time spent reworking mistakes, waiting for support, or diverted in a wrong repair approach is not included in recorded repair time.

f. Machine cost - The manufacturer's contractual cost per unit to the Government.

g. Parts cost - The manufacturer's stipulated contractual price to the Government at the time of failure.

4.5.2 Starting reliability tests. During the 200-hour operating reliability tests, a record shall be kept of the number of pull starts required to start the engine for each test run. Over the duration of the testing (approximately 400 start-ups), the average number of pull starts shall not exceed six. If the number of pull starts is found to be

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consistently excessive during the first 50 hours of testing, the manufacturer shall be notified and given the opportunity to improve starting performance. The operator will strictly follow the manufacturer's starting instructions during all starting reliability testing.

4.4.3 Maintenance tests. During the 200-hour reliability test, the machine shall have a maintenance ratio (MR) of not more than 0.05 (see 6.5). Time spent in the diagnosis of a failure is excluded; however, once the nature of the repair is known, the time required to accomplish the task (with necessary parts and tools on hand) is recorded as unscheduled maintenance time. All scheduled maintenance shall be in accordance with the maintenance manual with one possible exception. Inspection and/or cleaning of exhaust ports shall be in accordance with the provisions in the reliability testing; i.e., exhaust ports shall be cleaned after 100 hours of testing or a 25 percent buildup in carbon, whichever comes first. In addition to determining the MR, the following specific elapsed time shall be recorded to determine compliance with acceptable limits:

a. Removal of the muffler and exhaust pipe shall be 10 man-minutes or less.

b. Conversion of the machine from one application mode to another (i.e., liquid to granules, granules to dust) shall be 10 man-minutes or less.

All of the above operations as well as all scheduled maintenance (not unscheduled maintenance) shall be performed only with tools provided in the service kit.

4.5.4 Dispersal rate calibration tests. Calibration of the dispersal rates shall be performed to compare actual dispersal rates with the values provided in the operator's manual. The tests shall be conducted at three settings (low, medium, high) for liquids, dusts, and granules. The dispersants used in the testing shall be those referenced in 2.3. A deviation of more than 10 percent shall constitute failure of this test. During the calibration tests, the horizontal throw distance shall be measured to determine if it is a minimum of 20 feet for all three formulations.

4.5.5 Droplet size tests. Droplet size shall be determined by the conventional slide-wave technique, i.e., hand waving a Teflon coated microscope slide through the aerosol cloud three times in a vertical path with the coated side of the slide vertical to the ground and facing the aerosol cloud. The tests shall be conducted at a distance of 4 feet from the nozzle. The flow rate shall be 32 milliliters/minute or the nearest liquid flow setting. After measuring the diameter of a random sample of 200 droplets (approximately 400 power microscope is required), the volumetric median diameter (VMD) and the average large diameter (ALD) shall be

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calculated using a spread factor of 0.6. The VMD shall not exceed 40 microns and the ALD shall not exceed 200 microns. The ALD is defined as the average diameter of the 3 largest droplets measured in the sample of 200.

4.5.6 Air stream volume - velocity output test. The engine shall be set at the manufacturer's maximum recommended continuous operating speed. The average dispersal air stream velocity shall be measured with an anemometer to determine if it is 150 mph or greater.

4.5.7 Noise level test. Noise measurements shall be taken at points equal to the location of the ears of an operator. Noise levels shall be recorded when the unit is at maximum rated output. Data shall include dB(A) levels and octave band pressure levels and shall be in compliance with MIL-STD-1474.

4.6 Post test cleanliness. Upon completion of testing, each unit shall be emptied and flushed of all gasoline, spray, or dust and prepared for storage according to the manufacturer's standard procedure.

4.7 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the sprayer-duster shall be inspected to determine compliance with the requirements of section 5 of this specification.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A. The sprayers shall be preserved and packaged in accordance with level A of MIL-S-12651.

5.1.2 Level B. The sprayers shall be preserved and packaged in accordance with level B of MIL-S-12651.

5.2 Packing. The packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A. The sprayers shall be packed in accordance with level A of MIL-S-12651 as specified for boxed sprayers.

5.2.2 Level B. The sprayers shall be packed in accordance with level B of MIL-S-12651 as specified for boxed sprayers.

5.2.3 Commercial. The sprayers shall be packed in accordance with commercial requirements of MIL-S-12651.

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5.3 Marking. Marking shall be in accordance with the requirements of MIL-STD-129 and MIL-STD-130.

6. NOTES

6.1 Intended use. The sprayer-duster is intended primarily for spraying and dusting pesticides for the control of insect vectors of disease (mosquitoes, flies, ticks, and fleas). It can also be used to control economic Arthropod pests, soil infesting insects, and for herbicide applications.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Level of preservation and packaging if other than specified in section 5.

c. When a service kit is required (see 3.16).

d. Color of the finish coat of paint (see 3.15).

e. When a high mortality spare parts kit shall be required (see 3.20).

6.3 Qualification. With respect to products requiring certification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is the US Army Medical Bioengineering Research and Development Laboratory, Fort Detrick, Frederick, MD 21701-5010. Information pertaining to qualification of products may be obtained from this activity.

6.4 Revision changes. Asterisks are not used in the revision to identify changes with respect to the previous issue due to the extensive changes involved.

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6.5 Definitions. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total active operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded.

6.6 Noise limits. When the noise level of the sprayer exceeds 85 db(A), manuals must include the requirement for operator hearing protection; and equipment must be marked in accordance with MIL-STD-1474.

6.7 Provisions. The contracting officer should include provisioning requirements for repair parts and special tools, as necessary, and instructions on shipment of equipment.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Review Activities:

Army - MD, CE
Navy - MS, CG
DLA - CS

Preparing activity:

Army - ME

Project No. 3740-0116

Preparing activity agent:

US Army Medical Bioengineering
Research and Development Laboratory

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-23891C	Sprayer-Duster, Backpack Type; GED Blower (Commercial)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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